Universidad Nacional de Loja

Faculty of Energy, Industry, and Non-Renewable Natural Resources

Electrical Engineering Program

Design of a High-Voltage Transmission Line

Transmission Lines and Electrical Substations

AUTHORS:
Leonardo Gabriel Granda Erazo
Anthony José Espinoza Costa

INSTRUCTOR:
Eng. Jorge Isaac Peralta Córdova

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1. Introduction

The design of high-voltage transmission lines is a fundamental area of electrical engineering because it enables generation resources to be integrated safely and efficiently into the national interconnected power system. These lines transport large blocks of electric power from generating stations to load centers over long distances while maintaining acceptable losses. This study determines an economical voltage level for a 225-mile, three-phase transmission line supplying a 125-MVA load at unity power factor while maintaining nominal voltage at the receiving end.

The design problem is addressed from both technical and economic perspectives. The proposed system must satisfy energy-efficiency, voltage-regulation, electrical-safety, and structural-feasibility criteria while accounting for representative environmental conditions in Loja, Ecuador, including altitude and climate. The analysis applies the distributed-parameter long-line model, evaluates corona losses, and establishes practical criteria for selecting conductors, supporting structures, and insulation components.

The principal objective is to design a transmission line compatible with the voltage levels of Ecuador's National Interconnected System (SNI), while ensuring efficient, safe, and standards-oriented operation. The specific objectives are to determine the economical transmission voltage; select the number of circuits and a suitable conductor configuration; calculate the fundamental electrical parameters, including inductance, capacitance, and the generalized A, B, C, and D constants; evaluate voltage regulation, efficiency, and corona losses; and design an insulation system for normal operating and overvoltage conditions.

The scope is limited to the electrical design of the line. The distributed-parameter π representation is used exclusively, together with standard design data and the technical literature, particularly the transmission-line text by Luis María Checa. Specialized simulation software and detailed mechanical, civil, and environmental studies are outside the scope; the work instead develops an analytical design procedure that can be extended to practical projects.

2. Theoretical Framework

2.1 High-Voltage Power Transmission

High-voltage electric-power transmission provides several important technical advantages:

- Lower line losses, because these losses are inversely proportional to the square of the operating voltage.

- The ability to transmit electric power over long distances, thereby interconnecting remote generation facilities with urban load centers.
- High transmission efficiency for a given current density.
- A low percentage voltage drop, which improves voltage regulation.
- Lower current, which permits the use of smaller conductor cross sections and may reduce material and installation costs.

High-voltage transmission also entails the following disadvantages:

- More demanding maintenance procedures and safety measures because of the hazards associated with high electric-field strengths.
- Higher insulation requirements and, consequently, greater initial capital costs.
- Potentially significant visual and environmental impacts.
- More severe corona activity, which may cause power losses, audible noise, and electromagnetic interference.

2.2 Transmission-Line Parameters

In electric power systems, conductors constitute the physical medium through which energy is transported from generating stations to load centers. Proper selection of conductor material, construction, and configuration is essential to optimize both electrical performance and mechanical design. These characteristics directly affect line impedance, voltage drop, and energy efficiency.

Conductor materials According to [1], the materials most commonly used in transmission-line conductors are copper, aluminum, and aluminum conductor steel-reinforced (ACSR):

- **Copper (Cu):**
  Copper provides high electrical conductivity and corrosion resistance. Its comparatively high cost and mass, however, restrict its use in long overhead spans. It remains suitable where high current-carrying capability is required within a limited installation area.
- **Aluminum (Al):**
  Aluminum is lighter and less expensive than copper and provides approximately 60% of its conductivity. Its low mass makes it the most widely used material for medium- and high-voltage overhead transmission lines.
- **ACSR (Aluminum Conductor Steel Reinforced):**
  An ACSR conductor comprises aluminum strands surrounding a steel core. The steel core provides mechanical strength, whereas the aluminum strands carry most

of the current. This construction is extensively used for long spans and demanding environmental conditions.

Conductor construction Conductors may also be classified according to their physical construction:

- **Solid conductors:**
  These consist of a single conducting wire. Although they provide adequate conductivity, their rigidity limits their application to fixed installations and short spans.
- **Stranded conductors:**
  These consist of multiple interwoven strands, providing greater flexibility, vibration resistance, and ease of installation.
- **Hollow conductors:**
  These are mainly used when conductor mass must be reduced without an excessive reduction in conductivity. They are less common and are reserved for specialized applications.

Bundled conductors High- and extra-high-voltage systems commonly use bundled conductors, in which each phase comprises several subconductors separated by spacers. This configuration is typically adopted at voltage levels of 220 kV and above.

Advantages:

- Reduced corona activity, with lower electrical losses, audible noise, and electromagnetic emissions.
- Lower inductive reactance and improved active-power transfer capability.
- Greater current-carrying capacity without a substantial temperature increase.
- Reduced interference with nearby communication circuits.

Disadvantages:

- Higher installation cost because additional conductors, spacers, and structural components are required.
- More complex mechanical design because of spacer requirements and increased wind loading.
- More demanding maintenance because several subconductors must be inspected in each phase.

2.2.1 Transmission-Line Resistance

The electrical resistance of a conductor opposes current flow and produces energy losses as heat. It depends primarily on:

- Conductor material, such as copper or aluminum.
- Conductor length, with resistance increasing in proportion to length.
- Cross-sectional area, with resistance decreasing as area increases.
- Temperature, because the resistance of metallic conductors increases with temperature.

According to [1], resistance per unit length is a fundamental transmission-line modeling parameter and must be included when calculating Joule losses.

2.2.2 Transmission-Line Inductance

Inductance characterizes a conductor's opposition to changes in current through the magnetic field produced by that current. In a transmission line, each conductor induces an electromotive force in itself and in adjacent conductors. The principal factors that affect inductance are:

- Spacing between conductors.
- Conductor diameter.
- Geometric configuration of the line.

Inductance is particularly important in medium- and high-voltage lines because it contributes to total line impedance and therefore affects voltage regulation and reactive-power flow.

2.2.3 Transmission-Line Capacitance

Capacitance represents the ability of a line to store electric charge between conductors, between phases, and between each conductor and ground, thereby establishing an electric field. The principal factors that affect capacitance are:

- Spacing between conductors.
- Conductor height above ground.
- Conductor radius.
- Permittivity of the surrounding medium, which is air in an overhead line.

Capacitance is especially significant in long high-voltage lines because it may produce the Ferranti effect—an increase in receiving-end voltage under open-circuit or light-load conditions—and affects reactive-power distribution.

Resistance, inductance, and capacitance are fundamental line-model parameters and directly influence power-quality and operating performance [1].

2.3 Transmission-Line Transposition

Transposition is the periodic interchange of the physical positions occupied by the phase conductors along a three-phase overhead line. The line is divided into three sections of equal length so that each phase occupies every physical position for the same distance.

This procedure improves electromagnetic balance among the phases and reduces unequal voltage drops and asymmetrical line behavior. When phase conductors are arranged asymmetrically, their inductances and capacitances differ, producing:

- Unequal electromagnetic coupling among phases.
- Appreciable differences in apparent phase impedances.
- Unbalanced line constants.

Transposition also reduces voltages induced by electromagnetic and electrostatic coupling in communication lines that run parallel to power lines, thereby mitigating communication noise and related hazards. A complete sequence in which a conductor occupies all three positions is commonly referred to as a transposition cycle. Upon completion of a cycle, induced voltages are balanced and the average effects of mutual magnetic flux are substantially canceled.

Thus, line transposition balances the phase constants of three-phase systems with asymmetrical spacing and reduces the effects of mutual inductance [1].

2.4 Long Transmission Lines

A long transmission line is generally defined as one whose length exceeds 250 km and whose operating voltage exceeds 100 kV. Unlike short- and medium-line models, the electrical parameters R, L, C, and G are treated as uniformly distributed along the line. The rigorous solution is therefore based on differential equations that describe voltage and current as functions of position.

2.4.1 Propagation Constant

The propagation constant is a complex quantity that describes the attenuation and phase displacement of voltage and current waves along the line. It consequently characterizes both incident and reflected waves in a long transmission line.

2.4.2 Evaluation of Long-Line Constants

The total series impedance and shunt admittance are obtained from their corresponding per-unit-length parameters and the total line length. Common equivalent circuits include:

- Nominal-T model: the equivalent series impedance and shunt admittance are arranged in a T network and corrected according to line length and propagation behavior.
- Nominal-π model: the shunt admittance is divided into two equal components placed at the sending and receiving ends, with the corresponding hyperbolic correction.

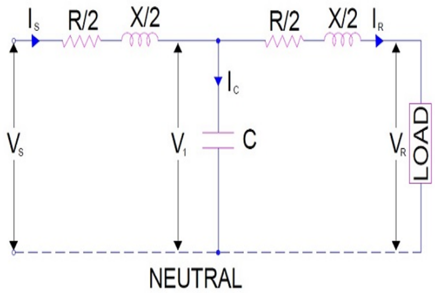

Figure 1. Nominal-T transmission-line model.

Source: Adapted from [2].

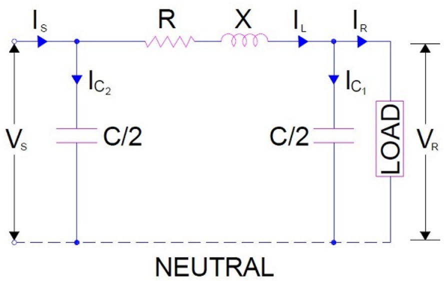

Figure 2. Nominal-π transmission-line model.

Source: Adapted from [2].

2.5 Transmission-Line Phenomena

In addition to their basic electrical parameters, transmission lines are subject to physical and electromagnetic phenomena that affect alternating-current operation. These include the **skin effect**, which changes the distribution of current within a conductor, and **charging current**, which results from distributed line capacitance even when no load is connected. These effects are relevant to the design, modeling, and operation of high-voltage and long-distance power systems.

2.5.1 Skin Effect

Direct current is distributed approximately uniformly over a conductor cross section. With alternating current, however, the current density becomes nonuniform and increases toward the conductor surface. This phenomenon is known as the *skin effect*.

Consequently, the effective AC resistance exceeds the DC resistance because current flows through a smaller effective cross-sectional area. The principal factors affecting skin effect are:

- Current frequency (higher frequency $\Rightarrow$ stronger effect).
- Conductor diameter (larger diameter $\Rightarrow$ stronger effect).
- Conductor material, particularly its permeability and resistivity.
- Conductor construction, with a generally lower effect in stranded conductors.

Skin effect is most significant at high frequencies and for large conductor cross sections. At power frequencies of 50–60 Hz, its effect is generally small but may not be negligible for large conductors [1].

2.5.2 Line Charging Current

In long transmission lines, distributed capacitance produces current even when the receiving end is unloaded. This **charging current**:

- Is in quadrature with the voltage, with an approximate phase displacement of 90°.
- Reaches its maximum value at the sending end.
- Decreases along the line toward the open receiving end.

Even when the receiving end is open, charging current circulates and produces active-power losses in the series resistance.

2.5.3 Ferranti Effect

The **Ferranti effect** occurs when the receiving-end voltage of a medium or long transmission line exceeds the sending-end voltage under no-load or light-load conditions.

The charging current produces a voltage component across the series reactance that causes $V_r > V_s$. The effect becomes more pronounced as line length increases and load decreases. A simplified estimate may be obtained by concentrating half of the total shunt capacitance at the receiving end and neglecting line resistance relative to inductive reactance.

2.5.4 Corona Effect

The **corona effect** is the ionization of air surrounding a conductor when the electric-field intensity exceeds a critical value, approximately 30 kV/cm peak or 21.1 kV/cm RMS under standard conditions. Corona may be identified by:

- A faint violet glow around the conductor.
- Audible hissing noise.
- Ozone formation.

Corona losses can be estimated using empirical expressions such as the Peek or Peterson equations. They depend on:

- Operating voltage and frequency.
- Atmospheric conditions, including pressure, temperature, rain, and snow.
- Conductor size, geometry, and surface condition.
- Spacing between conductors.

2.6 Overhead-Line Insulators

Insulators perform two essential functions in overhead transmission and distribution lines: they provide electrical isolation between energized conductors and supporting structures, and they withstand substantial mechanical loads. Because each insulator forms part of the conductor-support path, its failure may interrupt service.

2.6.1 Insulator Classification

According to their configuration and application, overhead-line insulators are principally classified as:

- **Pin-type insulators:** mounted directly on supporting structures and used mainly on low- and medium-voltage distribution lines.
- **Suspension insulators:** assembled as strings of series-connected units and used on high- and extra-high-voltage lines.
- **Strain insulators:** designed to withstand high mechanical tensile loads and typically installed at line terminations and angle structures.
- **Post insulators:** rigid units that provide both support and insulation and are commonly used in substations.

The most common insulating materials are:

- **Porcelain:** a mechanically robust ceramic with high dielectric strength.

- **Toughened glass:** a material that facilitates visual detection of damaged units.
- **Polymeric composites:** lightweight materials with good resistance to pollution and vandalism.

Each material provides distinct insulation, mechanical-strength, durability, and environmental-performance characteristics.

2.6.2 Overvoltage Performance

Insulators are exposed to severe electrical stress during lightning events and switching operations. Two critical failure mechanisms are:

- **Flashover:** an external disruptive discharge that occurs when the applied voltage exceeds the dielectric strength of the air path along the insulator surface.
- **Puncture:** an internal discharge through the insulating material that causes permanent damage.

To prevent these phenomena, insulators are specified with dry and wet flashover withstand levels above the maximum operating voltage and are tested in accordance with applicable international standards.

2.6.3 Insulator Strings

At voltage levels above approximately 33 kV, suspension-insulator strings comprising series-connected discs are commonly used. Voltage is not distributed uniformly because of parasitic capacitances. **Grading rings** are therefore used to redistribute the electric field and reduce excessive stress on the line-end units.

2.6.4 Voltage Distribution and Insulator-String Efficiency

An insulator string comprises multiple series-connected discs, each of which exhibits:

- **Mutual capacitance** (C) between adjacent metallic fittings.
- **Shunt capacitance** (mC) between each metallic fitting and ground through the tower structure.

These capacitances produce a nonuniform voltage distribution: the unit closest to the conductor is subjected to the highest voltage, whereas the unit nearest the crossarm is subjected to the lowest. The nonuniformity increases with the number of units in the string.

A string efficiency of 100% would indicate equal voltage across every unit. Although unattainable in practice, higher efficiency can be obtained by:

- Reducing (m), for example by increasing the distance from the string to the tower.
- Capacitance grading of the insulator units.
- Installing grading rings.

These measures produce a more uniform voltage distribution, reduce dielectric stress on the line-end unit, and improve line reliability.

3. Results

This chapter presents the results obtained by applying the design procedures proposed in Khanal [3] and Hona [4].

Table 1 summarizes the design data. The line must transmit 125 MVA over 225 miles while satisfying the specified performance criteria.

Table 1. System parameters

Parameter	**Value**
Length	225 mi = 362.025 km
Frequency	60 Hz
Number of circuits	1
Apparent power	125 MVA
Power factor	1
Active power	125 MW

The following sections apply the procedure established in the cited references.

3.1 Economical Voltage and Number of Circuits

The economical voltage is the transmission voltage that provides an appropriate compromise between power losses and infrastructure cost for a specified power-transfer requirement.

The empirical expression used to estimate the economical voltage is given in (1).

$$V_{eco} = 5.5 \cdot \sqrt{\frac{L}{1.6} + \frac{P \cdot 1000}{150 \cdot N_{circuitos} \cdot fp}} \tag{1}$$

For the initial calculation, a single circuit is assumed; therefore,

$$V_{eco} = 5.5 \cdot \sqrt{\frac{362.025}{1.6} + \frac{125 \cdot 1000}{150 \cdot 1 \cdot 1}}$$

$$V_{eco} = 179.03\ kV$$

The calculated value of V_{eco} does not coincide with a standardized voltage level. Because it exceeds 138 kV, the next higher standard level is adopted, yielding a selected transmission voltage of **230 kV**.

3.1.1 Number of Circuits

Following Hona [4], the required number of circuits is assessed using the parameter mf_{limit}. This correction factor relates line length to economical loading capability and decreases as transmission distance increases. The adopted values are listed in Table 2.

Table 2. Relationship between line length and mf_{limit}

Length (km)	$\boldsymbol{mf_{limit}}$
80	2.70
160	2.25
240	1.75
320	1.35
480	1.00
640	0.75

The applicable value of mf_{limit} is obtained by linear interpolation of Table 2 using (2).

$$y = y_1 + (x - x_1)\left(\frac{y_2 - y_1}{x_2 - x_1}\right) \tag{2}$$

Substitution of the design data gives:

$$y = 1.35 + (362.025 - 320)\left(\frac{1 - 1.35}{480 - 320}\right)$$

$$y = mf_{limit} = 1.258$$

The surge-impedance loading (SIL) must also be calculated. Equation (3) uses the characteristic impedance Z_c, taken as 400 Ω for one circuit and 200 Ω for two circuits.

$$SIL = \frac{V^2}{Z_c} \tag{3}$$

For each configuration, $mf_{calculated}$ is defined as the ratio of transmitted active power to the corresponding SIL. The results are listed in Table 3.

Table 3. Calculated SIL and $mf_{calculated}$

Circuits	SIL	$\boldsymbol{mf_{calculated}}$
1	132.25	0.9451
2	264.50	0.4725

An acceptable configuration must satisfy $mf_{calculated} \leq mf_{limit}$. Both alternatives meet this requirement, although the double-circuit case provides a larger margin. The single-circuit configuration is retained because it requires less infrastructure and is therefore expected to have a lower capital cost.

3.2 Conductor Selection

The conductor is selected to provide adequate ampacity while limiting losses sufficiently to achieve an efficiency greater than 94% [3].

ACSR conductors are evaluated using the data reported by Grainger and Stevenson [5] and Conductores Monterrey S.A. de C.V. [6] and the current, loss, and efficiency expressions in (4)–(6).

$$I = \frac{P}{\sqrt{3} \cdot V_{LL} \cdot fp \cdot N_{Circuitos}} \tag{4}$$

$$P_{Losses} = 3 \cdot I^2 \cdot R_{\Omega/km} \cdot length \tag{5}$$

$$\eta = \frac{P_{out}}{P_{in}} \cdot 100\% = \frac{P_{out}}{P_{out} + P_{Losses}} \cdot 100\% \tag{6}$$

Equation (4) gives a load current of **0.3137 kA**. Candidate conductors from OSPREY (282 mm^2) through FALCON (805.7 mm^2) are evaluated to identify an ampacity-compliant option with low resistive losses, as shown in Fig. 3.

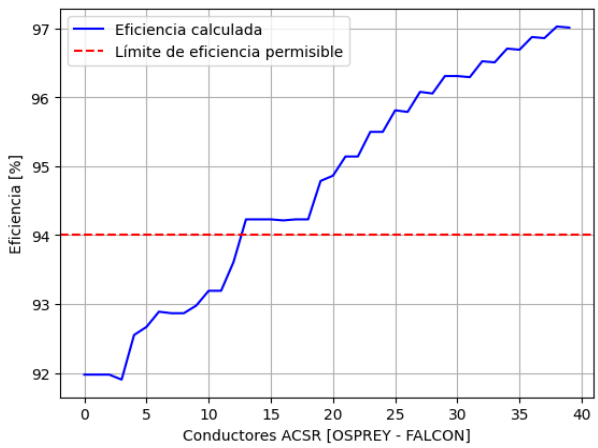

Figure 3. Calculated efficiency for the candidate conductors.

Source: Prepared by the authors.

The theoretical efficiency exceeds 94% from approximately candidate 13 (STARLING) onward. The ACSR FINCH conductor is selected from this acceptable range; its principal characteristics are listed in Table 4.

Table 4. Characteristics of the ACSR FINCH conductor

Characteristic	Value
Designation	1113 $AWG/kcmil$
Nominal overall diameter	32.85 mm
Cross-sectional area	564 mm^2
Current-carrying capacity	1100 A
Resistance	0.0514 $\Omega_{DC-20°C}$
RMG / D_s	0.0436 ft

After selecting the conductor, (5) and (6) give:

$$P_{Losses} = 3 \cdot (0.3137)^2 \cdot 0.0514 \cdot 362.025$$

$$P_{Losses} = 5.4962\ MW$$

$$\eta = \frac{125}{125 + 5.4962} \cdot 100\%$$

$$\eta = 95.788\%$$

These values are theoretical. Actual performance also depends on ambient conditions, conductor condition, temperature, loading history, and other operational variables that are outside the scope of this study.

3.3 Tower Selection

Detailed structural design is outside the scope of this study. A representative tower configuration is therefore selected from the structures presented by Checa [7].

Because the reference does not provide a specific 230-kV arrangement, the closest available configuration, rated at 220 kV, is adopted. It is shown in Fig. 4.

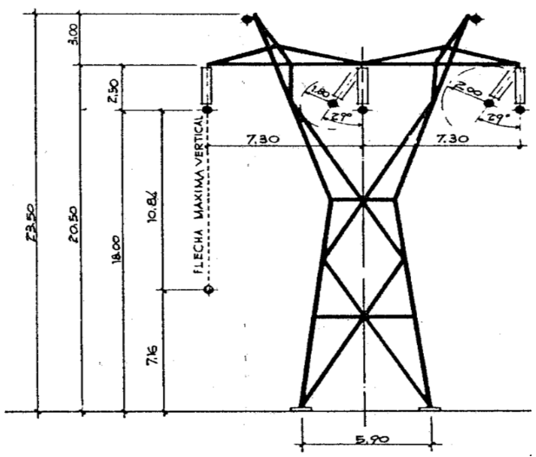

Figure 4. Single-circuit 220-kV line configuration.

Source: Adapted from [7].

The tower geometry is used to calculate the geometric mean distance (GMD), which represents the equivalent phase spacing. For a single-circuit, three-phase line,

$$DMG = \sqrt[3]{D_{AB} \cdot D_{BC} \cdot D_{CA}} \tag{7}$$

The resulting GMD is **9.197 m**.

3.4 Calculation of Line Parameters

3.4.1 Line Inductance and Inductive Reactance

Alternating current establishes a time-varying magnetic flux around the conductor. Inductance, measured in henries, is defined as flux linkage per unit current.

Accounting for the internal and external flux linkages yields the per-unit-length expression in (8).

$$L_{Line} = 2 \cdot 10^{-7} \cdot \ln \frac{DMG}{RMG_L} \cdot 1000 \tag{8}$$

For impedance calculations, the corresponding inductive reactance is obtained from line length, inductance per unit length, and system frequency:

$$X_L = 2\pi \cdot frequency \cdot L_{Line} \cdot length \tag{9}$$

3.4.2 Line Capacitance and Susceptance

Any two conductors separated by a dielectric constitute a capacitor. In an overhead line, air is the principal dielectric; consequently, capacitance exists between phases and between each phase and neutral.

Capacitance is defined as stored charge per unit potential difference.

The per-unit-length line capacitance is given by (10). Its denominator is analogous to that of the inductance expression, except that the capacitive geometric mean radius is equal to the physical conductor radius.

$$C_{Line} = \frac{2\pi \cdot 8.854 \cdot 10^{-12}}{\ln \frac{DMG}{RMG_C}} \cdot 1000 \tag{10}$$

After calculating phase-to-neutral capacitance, the corresponding capacitive reactance is obtained from (11). With capacitance expressed in F/km, consistent line-length units must be used.

$$X_C = \frac{1}{2\pi \cdot frequency \cdot C_{Line} \cdot length} \tag{11}$$

The shunt susceptance contributed by the line capacitance is the reciprocal of the capacitive reactance:

$$Y_C = B_C = \frac{1}{X_C} \tag{12}$$

3.4.3 Line Impedance and Admittance

The series impedance is the complex sum of conductor resistance and inductive reactance, both expressed in ohms:

$$Z_{Line} = R_{[\Omega]} + jX_{L[\Omega]} \tag{13}$$

The shunt admittance comprises conductance and susceptance. Because leakage conductance is neglected in this study, the line admittance is represented solely by the capacitive susceptance:

$$Y_{Line} = jB_{C[S]} \tag{14}$$

3.4.4 Line ABCD Parameters

Medium and long lines are represented by the generalized circuit constants A, B, C, and D. For a distributed-parameter long line, these constants depend on characteristic impedance and propagation constant.

The characteristic impedance and the propagation constant multiplied by line length are calculated from (15) and (16), respectively.

$$Z_c = \sqrt{\frac{|z_{[\Omega/km]}|}{|y_{[S/km]}|}} < \frac{\theta_z - \theta_y}{2} \tag{15}$$

$$\lambda l = length \cdot \sqrt{|z_{[\Omega/km]}| \cdot |y_{[S/km]}|} < \frac{\theta_z + \theta_y}{2} = \alpha + j\beta \tag{16}$$

Because both quantities are complex, their rectangular and polar forms must be handled consistently. The long-line equations require hyperbolic functions of a complex argument.

With α and β denoting the real and imaginary components of λl, the required functions are evaluated in radians as follows:

$$\sinh_{\lambda l} = \sinh(\alpha)\cos(\beta) + j\cosh(\alpha)\sin(\beta) \tag{17}$$

$$\cosh_{\lambda l} = \cosh(\alpha)\cos(\beta) + j\sinh(\alpha)\sin(\beta) \tag{18}$$

The generalized line constants are then obtained from (19)–(22).

$$A = \cosh \lambda l \tag{19}$$

$$B = Z_c \sinh \lambda l \tag{20}$$

$$C = \frac{\sinh \lambda l}{Z_c} \tag{21}$$

$$D = A = \cosh \lambda l \tag{22}$$

Evaluation with the specified design data yields Table 5.

Table 5. Electrical parameters of the 230-kV simplex line

Parameter	**Magnitude**	**Angle (°)**
Line inductance, L	1.30795 mH/km	—
Line capacitance, C	8.79114 nF/km	—
Inductive reactance, X_L	178.5098 Ω	90
Capacitive reactance, X_C	833.46 Ω	-90
Susceptance, B_C	1199.8168 μS	90
Parameter A	0.8948	0.6895
Parameter B	173.1391	84.2652
Parameter C	0.0011574	90.2163
Parameter D	0.8948527	0.6895

3.5 Line-Performance Calculation

3.5.1 Initial Calculation: Simplex Configuration

Once the $ABCD$ parameters are known, the sending-end phase voltage and current can be calculated from the receiving-end quantities using the two-port relationship in (23).

$$\begin{bmatrix} V_{\phi S} \\ I_S \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_{\phi R} \\ I_R \end{bmatrix} \tag{23}$$

The sending-end complex power, required to evaluate line efficiency, is calculated from (24).

$$S_s = 3 \cdot V_{S\phi} \cdot I_L^* = P_S + jQ_S \tag{24}$$

Efficiency follows from (6) once the sending-end active power has been obtained. Voltage regulation is evaluated using (25), which is applicable when the appropriate line constants have been determined.

$$\%R_V = \frac{\frac{|V_{LS}|}{|A|} - |V_{LR}|}{|V_{LR}|} \cdot 100\% \tag{25}$$

The calculated simplex-line performance is summarized in Table 6.

Table 6. Performance of the 230-kV simplex line

Quantity	**Magnitude**	**Unit**
Sending-end phase voltage, $V_{S\phi}$	136.0741 ∠ 24.064	kV
Sending-end line voltage, V_{SL}	235.6873 ∠ 54.064	kV
Sending-end current, I_S	0.3212 ∠ 29.2757	kA
Voltage regulation, $\%R_V$	14.5135	%
Sending-end active power, P_S	130.583	MW
Sending-end reactive power, Q_S	-11.91	MVAr
Efficiency, η	95.72	%

The design criterion adopted from Khanal [3] requires voltage regulation below 12%. The simplex configuration does not satisfy this condition and must therefore be revised. Its efficiency of 95.72% does exceed the 94% requirement and closely agrees with the preliminary resistance-only estimate.

3.5.2 Recalculation with Two-Conductor Bundles

Bundled conductors reduce line reactance and can improve voltage regulation. A duplex configuration using the same conductor type is therefore evaluated, together with the compatible tower arrangement from Checa [7] shown in Fig. 5.

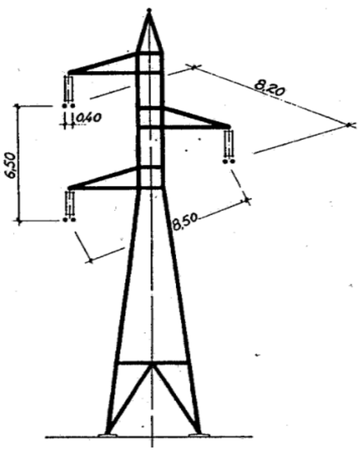

Figure 5. Duplex 220-kV line configuration.

Source: Adapted from [7].

The revised tower geometry changes the GMD and the inductive and capacitive geometric mean radii (GMRs), which in turn modify impedance, admittance, $ABCD$ parameters, and line performance.

Application of (7) to the new geometry gives a GMD of **7.68 m**.

For a two-conductor bundle, the equivalent inductive GMR is calculated from the simplified expression reported by Grainger and Stevenson [5]:

$$RMG_L^2 = \sqrt{RMG_L \cdot s} = \sqrt{D_s \cdot s} \tag{26}$$

The capacitive GMR is calculated similarly, using the physical conductor radius:

$$RMG_C^2 = \sqrt{RMG_C \cdot s} = \sqrt{r \cdot s} \tag{27}$$

Here, s is the subconductor spacing, equal to 40 cm.

Using these equivalent radii, (8)–(25) yield the parameters and performance reported in Tables 7 and 8.

Table 7. Electrical parameters of the 230-kV duplex line

Parameter	**Magnitude**	**Angle (°)**
Line inductance, L	0.9314 mH/km	—
Line capacitance, C	12.22 nF/km	—
Inductive reactance, X_L	127.1241 Ω	90
Capacitive reactance, X_C	599.45502 Ω	-90
Susceptance, B_C	1668.1818 μS	90
Parameter A	0.89591	0.9579
Parameter B	123.9863	81.973
Parameter C	0.0016098	90.3007
Parameter D	0.89591	0.9579

Table 8. Performance of the 230-kV duplex line

Quantity	**Magnitude**	**Unit**
Sending-end phase voltage, $V_{S\phi}$	130.81583 ∠ 18.0403	kV
Sending-end line voltage, V_{SL}	226.5796 ∠ 48.04033	kV
Sending-end current, I_S	0.35511 ∠ 37.9674	kA
Voltage regulation, $\%R_V$	9.95805	%
Sending-end active power, P_S	131.0187	MW
Sending-end reactive power, Q_S	-47.4983	MVAr
Efficiency, η	95.4062	%

The duplex configuration yields a voltage regulation of **9.958%**, which satisfies the adopted limit. The modified line parameters increase sending-end current and losses, reducing efficiency to **95.4%**; nevertheless, this value remains acceptable.

A triplex configuration is also evaluated using the arrangement shown in Fig. 6.

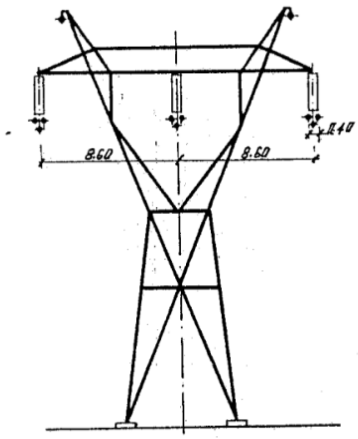

Figure 6. Triplex 220-kV line configuration.

Source: Adapted from [7].

The resulting voltage regulation and efficiency are compared with those of the simplex and duplex cases in Figs. 7 and 8.

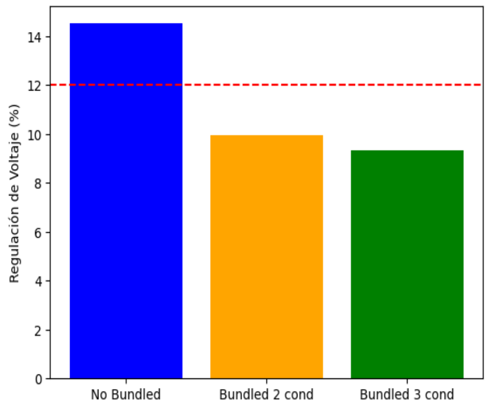

Figure 7. Voltage regulation for each conductor configuration.

Source: Prepared by the authors.

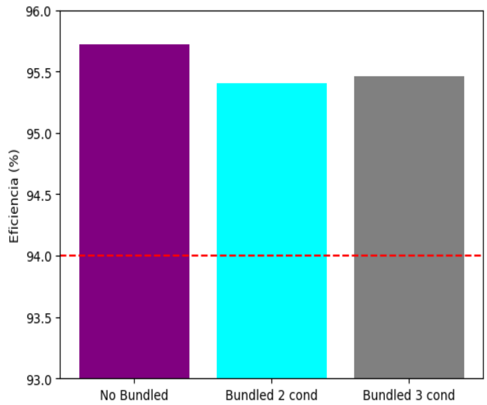

Figure 8. Line efficiency for each conductor configuration.

Source: Prepared by the authors.

3.6 Corona-Effect Calculation

Corona can produce power loss, audible noise, and radio interference. If the disruptive critical voltage exceeds the operating phase voltage, sustained corona losses are not expected under the assumed conditions.

The critical corona voltage is estimated from (28).

$$V_c = m_o \cdot g_o \cdot \delta \cdot r \cdot \ln \frac{DMG}{r} \tag{28}$$

The parameters are defined as follows:

- m_o is the conductor-surface irregularity factor $(0.8 - -1.0)$;
- g_o is the dielectric strength of air $\left(30/\sqrt{2}\ \mathrm{kV_{RMS}/cm}\right)$;
- δ is the air-density correction factor;
- r is the conductor radius in centimeters; and
- DMG is the geometric mean distance between conductors in centimeters.

The standard value 21.21 $\mathrm{kV_{RMS}/cm}$ applies at 76 cmHg and 25°C. For other atmospheric conditions, the correction factor is estimated from (29), where b is atmospheric pressure in cmHg and t is ambient temperature in degrees Celsius.

$$\delta = \frac{3.92 \cdot b}{273 + t} \tag{29}$$

Corona loss in kW/km/phase is estimated from (30).

$$P_{Loss}^{Corona} = 242.2 \cdot \left(\frac{frequency + 25}{\delta} \right) \cdot \sqrt{\frac{r}{DMG}} \cdot (V_\phi - V_c)^2 \cdot 10^{-5} \tag{30}$$

For the assumed installation in Loja, an atmospheric pressure of 1019 hPa, equivalent to approximately **76.43 cmHg**, is adopted for the calculation.

Because temperature varies seasonally, an average value is used. The maximum and minimum temperature data shown in Fig. 9 yield an approximate mean of **18.3°C**.

For a stranded conductor, the recommended irregularity factor lies between 0.80 and 0.87; a value of $m_o = \mathbf{0.82}$ is adopted.

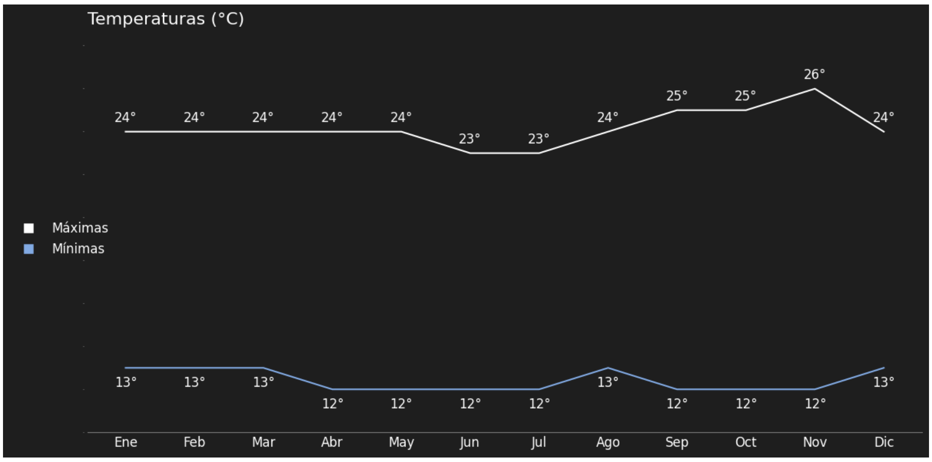

Figure 9. Temperature data for Loja, Ecuador.

Source: Data from National Centers for Environmental Information (NCEI) [8].

Substitution into (28) gives:

$$V_c = 180.62\ kV_{RMS}/\phi$$

$$V_c = 312.84\ kV_{RMS}$$

The calculated line-to-line critical voltage exceeds the nominal system voltage; therefore, corona loss is not expected under the assumed atmospheric and conductor-surface conditions.

3.7 Number of Insulator Units

The number of suspension-insulator units is determined by evaluating the following withstand conditions:

- One-minute dry withstand test.
- One-minute wet withstand test.
- Temporary-overvoltage withstand test.
- Lightning-impulse withstand test.
- Switching-impulse withstand test.

The expressions adopted from Hona [4] are given in (31)–(35).

$$V = 1\ min\ dry\ withstand\ voltage\ \cdot FWR \cdot NAC \cdot FS \tag{31}$$

$$V = 1\ min\ wet\ withstand\ voltage\ \cdot FWR \cdot NAC \cdot FS \tag{32}$$

$$OV\ \ FOV = EV \cdot Max\ system\ voltage \cdot \sqrt{2} \cdot FWR \cdot NAC \cdot FS \tag{33}$$

$$Lightning\ FOV = Impulse\ withstand \cdot FWR \cdot NAC \cdot FS \tag{34}$$

$$Switch\ FOV = \left(\frac{V_{System}}{\sqrt{3}} \cdot \sqrt{2} \cdot 1.1\right) \cdot SSR \cdot SIR \cdot FWR \cdot NAC \cdot FS \tag{35}$$

where:

- FWR is the flashover ratio (1.15);
- NAC is the nonstandard-atmosphere correction factor (1.1);
- FS is the safety factor (1.2);
- EV converts system voltage to temporary overvoltage (0.8);
- SSR is the switching-surge factor (2.8); and
- SIR is the switching-impulse ratio (1.2).

The remaining withstand and flashover values are obtained from the reference data reproduced in Figs. 10 and 11.

Max System Voltage (kV)	1 min. dry withstand (kV)	1 min. wet withstand (kV)	Impulse withstand (kV)
123	215	185	450
145	265	230	550
245	435	395	900
420	760	680	1550

Figure 10. Reference voltage-withstand capability.

Source: Adapted from Hona [4].

No. of discs	1 min. dry FOV (kV)	1 min. wet FOV (kV)	Impulse withstand (kV)
1	80	50	150
2	115	90	225
3	215	130	335
4	270	170	440
5	325	210	525
6	380	250	610
7	435	290	695
8	486	330	780
9	535	370	860
10	585	410	945
11	635	450	1025
12	685	485	1105
13	730	520	1185
14	775	565	1265
15	820	590	1345
16	865	620	1425

Figure 11. Reference flashover voltage.

Source: Adapted from Hona [4].

The maximum system voltage is taken as 1.05 p.u., equivalent to 241.5 kV. This value is used to select the applicable reference withstand levels.

The resulting requirements are summarized in Table 9.

Table 9. Calculated number of insulator units

Test condition	Voltage	Number of units
Dry withstand	660.33 kV	12 units
Wet withstand	599.61 kV	16 units
Temporary overvoltage	420.768 kV	11 units
Lightning impulse	1366.2 kV	16 units
Switching impulse	1053.62 kV	12 units

The governing requirement is therefore **16** insulator units.

3.8 Voltage Distribution Along the Insulator String

3.8.1 String Without a Grading Ring

Figure 12 shows the mutual capacitances (C) and shunt capacitances (mC) used in the equivalent circuit.

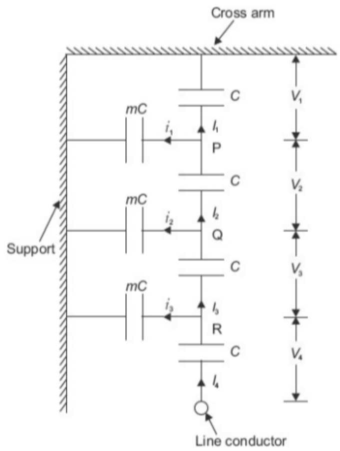

Figure 12. Equivalent circuit of an insulator string.

Source: Adapted from Sivanagaraju [1].

The shunt capacitances make the voltage distribution nonuniform, particularly for strings with many units. The empirical expression proposed by Sivanagaraju [1] for the voltage across each disc is given in (36).

$$V_{\mathrm{i}} = \frac{V_\phi \cdot 2\sinh\left(0.5\sqrt{m}\right)\cdot\cosh\left((n-\mathrm{i}+0.5)\sqrt{m}\right)}{\sinh n\sqrt{m}} \tag{36}$$

Here, m is the ratio of shunt capacitance to mutual capacitance, n is the number of units in the string, and i identifies the unit under evaluation.

Before calculation, a value of m must be selected. Fernández et al. [9] reports that the stray capacitance between metallic fittings and the grounded tower is of the order of 6 pF for representative high-voltage geometries and is therefore small in absolute terms.

The sources Poriyaan Engineering [10], Sivanagaraju [1], and Electrical Engineering Portal [11] indicate typical values of m between 0.1 and 0.2. A value closer to zero produces a more uniform voltage distribution. Although m can be reduced by increasing the string-to-tower clearance or crossarm length, structural and economic constraints limit this approach.

Accordingly, the lower representative value $\mathbf{m = 0.1}$ is selected.

Application of (36) gives the distribution in Table 10.

Table 10. Disc-voltage distribution: empirical analysis

Unit	Voltage (kV)
V_1	0.5421
V_2	0.5967
V_3	0.7116
V_4	0.8981
V_5	1.1753
V_6	1.5709
V_7	2.1250
V_8	2.8934
V_9	3.9535
V_{10}	5.4122
V_{11}	7.4167
V_{12}	10.1691
V_{13}	13.9469
V_{14}	19.1310
V_{15}	26.2442
V_{16}	36.0038
V_{Total}	132.7905

The distribution is strongly nonuniform, and the line-end unit is subjected to the highest voltage. String efficiency is calculated from (37).

$$\eta_{aislador} = \frac{V_\phi}{n \cdot V_n} \cdot 100\% \tag{37}$$

Maximum efficiency would require approximately equal voltage across all units. The denominator uses the maximum unit voltage, normally that of the line-end disc.

For the preceding distribution, the calculated efficiency is **25.0514**%, confirming severe voltage nonuniformity.

A second method applies Kirchhoff's current law directly to the equivalent circuit in Fig. 12. For a four-unit string, the recurrence begins as follows:

$$V_2 = V_1 + V_1 \cdot m$$

$$V_3 = V_2 + (V_1 + V_2)\, m$$

$$V_4 = V_3 + (V_1 + V_2 + V_3)\, m$$

The pattern yields the general recurrence in (38): the voltage across unit $\mathrm{i}+1$ equals that across unit i plus m times the cumulative voltage across the preceding units.

$$V_{\mathrm{i}+1} = V_{\mathrm{i}} + m\left(\sum_{j=1}^{\mathrm{i}} V_j\right), \quad \mathrm{i} \in \mathbb{N} \tag{38}$$

The sum of the individual unit voltages must equal the line phase voltage, as expressed in (39).

$$V_{T\phi} = \sum_{\mathrm{i}=1}^{N} V_{\mathrm{i}} \tag{39}$$

This recurrence produces the results in Table 11.

Table 11. Disc-voltage distribution: recurrence analysis

Unit	Voltage (kV)
V_1	0.5511
V_2	0.6062
V_3	0.7220
V_4	0.9099
V_5	1.1888
V_6	1.5867
V_7	2.1431
V_8	2.9139
V_9	3.9761
V_{10}	5.4359
V_{11}	7.4393
V_{12}	10.1866
V_{13}	13.9526
V_{14}	19.1139
V_{15}	26.1865
V_{16}	35.8778
V_{Total}	132.7906

The two methods produce closely comparable, although not identical, results. Because the recurrence follows directly from circuit laws and is straightforward to implement, it is used in the subsequent analysis.

Equation (37) gives an efficiency of 23.13%; thus, the distribution remains poor even with the minimum adopted value of m.

A grading ring is therefore evaluated as a practical means of improving voltage distribution.

3.8.2 String with a Grading Ring

A grading ring introduces additional capacitance between the metallic fittings and the energized conductor, thereby improving the voltage distribution. Although a perfectly uniform distribution is not attainable in practice, the improvement can be substantial. Figure 13 shows the modified equivalent circuit.

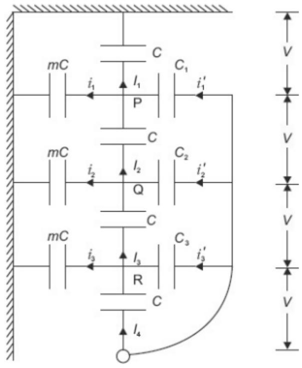

Figure 13. Equivalent circuit of an insulator string with a grading ring.

Source: Adapted from Sivanagaraju [1].

For this circuit, the voltage distribution is derived by applying Kirchhoff's laws rather than an empirical expression.

The ring capacitance is represented as the disc capacitance multiplied by a factor k. In the absence of a geometry-specific value, k is assumed equal to m, giving $k = 0.1$ for this case study.

Applying current balance at successive nodes gives:

$$V_2 = V_1 + V_1 m - (V_{T\phi} - V_1)\,k$$

$$V_3 = V_2 + (V_1 + V_2)\,m - [V_{T\phi} - (V_1 + V_2)]\,k$$

$$V_4 = V_3 + (V_1 + V_2 + V_3)\,m - [V_{T\phi} - (V_1 + V_2 + V_3)]\,k$$

The resulting general recurrence is given in (40).

$$V_{\mathrm{i}+1} = V_{\mathrm{i}} + m\left(\sum_{j=1}^{\mathrm{i}} V_j\right) - k\left[V_{T\phi} - \sum_{j=1}^{\mathrm{i}} V_j\right], \quad \mathrm{i} \in \mathbb{N} \tag{40}$$

Equations (39) and (40) produce the results in Table 12.

Table 12. Disc-voltage distribution with a grading ring

Unit	Voltage (kV)
V_1	23.8370
V_2	15.3254
V_3	9.8788
V_4	6.4079
V_5	4.2187
V_6	2.8732
V_7	2.1023
V_8	1.7520
V_9	1.7520
V_{10}	2.1023
V_{11}	2.8732
V_{12}	4.2187
V_{13}	6.4079
V_{14}	9.8788
V_{15}	15.3254
V_{16}	23.8370
V_{Total}	132.7906

Relative to Table 11, the grading ring substantially redistributes voltage, particularly across the tower-end units, and limits the maximum unit voltage to approximately 24 kV. Figure 14 compares the corresponding string efficiencies.

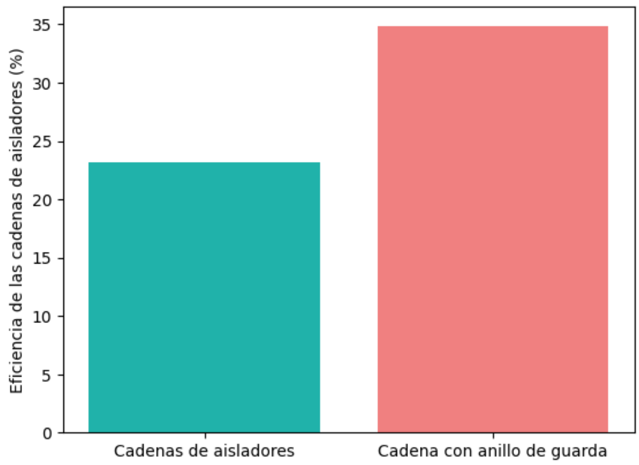

Figure 14. String efficiency with and without a grading ring.
Source: Prepared by the authors.

A sensitivity analysis varies one capacitance factor while holding the other at 0.1; the results are shown in Fig. 15.

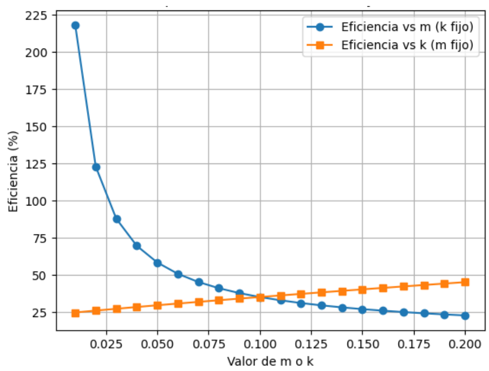

Figure 15. String efficiency for different values of m and k.

Source: Prepared by the authors.

The curves intersect at $m = k = 0.1$. Although further reducing m would improve efficiency, doing so would require clearances or structural dimensions that may be physically and economically impractical.

Increasing k improves efficiency, confirming that the additional capacitance introduced by the grading ring promotes a more uniform voltage distribution.

The calculated efficiencies remain moderate because uniform distribution is difficult to achieve across a 16-unit string. Shorter strings would generally exhibit greater sensitivity and higher efficiency for comparable parameter changes.

Finally, (38) and (40) are compared while holding $k = 0.1$ and varying the shunt-capacitance ratio. Figure 16 presents the result.

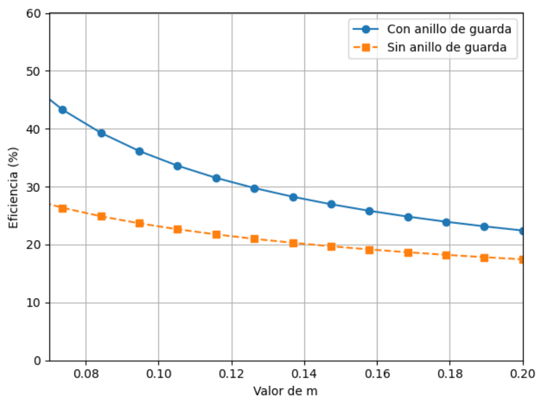

Figure 16. String efficiency with and without a grading ring.

Source: Prepared by the authors.

The comparison confirms that a grading ring is a technically effective means of improving unit-voltage distribution and limiting dielectric overstress.

3.8.3 Validation Using Commercial Insulator Units

The calculated voltages for the configurations with and without a grading ring are compared with the ratings of commercially available suspension units.

For the configuration without a grading ring, the N-160 toughened-glass suspension unit, catalog code 310961 CAT N-12 and CFE designation 25SVC111, is selected from [12]. Its ratings are listed in Table 13.

Table 13. Characteristics of the N-12 insulator unit

Description	Value
Dry flashover voltage at 60 Hz (kV)	80
Wet flashover voltage at 60 Hz (kV)	50
Positive critical impulse flashover voltage (kV)	125
Negative critical impulse flashover voltage (kV)	130
Radio-interference test voltage at 60 Hz (kV)	10
Maximum radio-interference voltage at 1 MHz (μV)	50
Low-frequency puncture voltage (kV)	110
Mechanical strength (kN)	111
Impact strength (N m)	7
Three-second routine mechanical load (kN)	55.5
Nominal diameter (mm)	258
Unit spacing (mm)	146
Minimum creepage distance (mm)	320
Approximate net mass (kg)	4.0

The unit has a 50-kV wet flashover rating, whereas the maximum calculated unit voltages in Tables 10 and 11 are 36.00 and 35.88 kV, respectively. It therefore provides adequate dielectric margin for the ungraded case under the adopted comparison criterion.

For the configuration with a grading ring, the N-160 glass unit with catalog code 310950 CAT CT-4 and CFE designation 17SVH044 is evaluated. Its ratings are given in Table 14.

Table 14. Characteristics of the CT-4 insulator unit

Description	**Value**
Dry flashover voltage at 60 Hz (kV)	60
Wet flashover voltage at 60 Hz (kV)	30
Positive critical impulse flashover voltage (kV)	100
Negative critical impulse flashover voltage (kV)	100
Radio-interference test voltage at 60 Hz (kV)	7.5
Maximum radio-interference voltage at 1 MHz (μV)	50
Low-frequency puncture voltage (kV)	80
Mechanical strength (kN)	44
Impact strength (N m)	5
Three-second routine mechanical load (kN)	22
Nominal diameter (mm)	175
Unit spacing (mm)	140
Minimum creepage distance (mm)	178
Approximate net mass (kg)	2.0

With the grading ring, the maximum calculated unit voltage is 23.83 kV (Table 12). Relative to the 30-kV wet flashover rating, the CT-4 provides a dielectric margin of approximately 6.2 kV and is therefore acceptable for the proposed case study under the adopted criterion.

The manufacturer reports compliance with CFE 52210-02, NMX-J-245-ANCE, ANSI C29.1–C29.2, and IEC 60120, 60305, and 60383-1.

The graded configuration provides the better voltage distribution and permits a smaller insulator unit for the same nominal line voltage.

The ring itself introduces additional cost, but the ungraded case requires a larger unit because of its higher electrical stress. The N-12 has a nominal diameter of 258 mm, whereas the CT-4 diameter is 175 mm; the latter is therefore expected to reduce the size and mass of the string assembly.

The N-12 is not technically unsuitable; it simply provides a larger withstand margin than required by this simplified comparison. Final insulation selection should be based on a lifecycle cost-benefit assessment that includes electrical, mechanical, pollution, altitude, maintenance, and procurement requirements.

Figure 17 shows a representative suspension unit.

Figure 17. Representative toughened-glass suspension-insulator unit.
Source: Adapted from [12].

4. Conclusions

The electrical design of the 225-mile, three-phase transmission line for a 125-MVA load is technically feasible and achieves an efficiency above 94%, maintains the specified receiving-end voltage, and provides acceptable voltage regulation.

The initial simplex configuration, with one conductor per phase, satisfied the efficiency criterion but exceeded the voltage-regulation limit adopted from [3] and [4]. The line was therefore reformulated as a duplex bundle with two subconductors per phase. This configuration improved voltage regulation at the expense of a modest reduction in efficiency.

Application of the distributed-parameter long-line model was essential for accurately determining the line inductance, capacitance, and generalized A, B, C, and D constants. The resulting model enabled a rigorous assessment of line behavior and produced an efficiency consistent with the preliminary estimate based solely on conductor resistance.

Conductor selection was governed primarily by energy-efficiency and current-carrying criteria. Efficiencies were calculated for a range of ACSR conductors to identify the sizes capable of exceeding the design threshold. These values are theoretical approximations because actual line performance also depends on environmental conditions, conductor condition, and operating constraints.

Insulation-system dimensioning was the most demanding component of the design. The analysis shows that a grading ring substantially improves the voltage distribution across a long insulator string. By limiting the concentration of dielectric stress on individual units under both normal and overvoltage conditions, the ring improves insulation reliability, permits the use of smaller commercial units in the evaluated case, and may extend service life.

5. Recommendations

- Conduct a comprehensive literature review before detailed design to establish the theoretical basis, applicable constraints, and technical justification for each design decision.
- Extend the analysis beyond the assumed unity-power-factor load. Future studies should evaluate several load levels and power factors so that the selected design remains satisfactory over the expected operating range.
- Complement the electrical design with mechanical, geotechnical, topographic, economic-feasibility, environmental, right-of-way, and safety studies before considering practical implementation.
- Replace the simplified climatic and atmospheric assumptions used in this academic case study with site-specific, quality-controlled measurements in a practical project.
- Automate repetitive calculations in a validated computational workflow to reduce development time, improve traceability, and minimize numerical errors. Python and Visual Studio Code were used in this study.
- Select conductor type and cross-sectional area by considering normal loading, emergency loading, thermal limits, sag, and Joule losses. Insulator selection should account for phase voltage, switching and lightning overvoltages, contamination, altitude, and other environmental correction factors.
- Periodically reassess emerging transmission technologies, including high-temperature low-sag conductors, corona and grading rings, flexible AC transmission system (FACTS) devices, and the operational implications of distributed generation.

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